

Multiple Choice

1. **Answer:** A

Options: A) 3.74 T; B) 5.19 T; C) 6.08 T; D) 9.49 T

Note: Correct option: A - 3.74 T

2. **Answer:** C

Options: A) all molecular bonds absorb IR radiation; B) IR peak intensities are related to molecular mass; C) most organic functional groups absorb in a characteristic region of the IR spectrum; D) each element absorbs at a characteristic wavelength

Note: Correct option: C - most organic functional groups absorb in a characteristic region of the IR spectrum

3. **Answer:** A

Options: A) HCl; B) AgCl; C) CaCl₂; D) CCl₄

Note: Correct option: A - HCl

4. **Answer:** A

Options: A) 17 O; B) 19 F; C) 29 Si; D) 31 P

Note: Correct option: A - 17 O

5. **Answer:** C

Options: A) K⁺; B) Ca₂⁺; C) Sc₃⁺; D) Rb⁺

Note: Correct option: C - Sc₃⁺

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Most organic compounds do not contain silicon atoms.'

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It has little effect.'

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Oxide'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 820. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 72.0 s. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key